Could missingness, duplicate rows, or recording practices distort our conclusions?

The quality audit found no missing values in the cleaned dataset and one repeated row beyond the first identical occurrence. No constant or entirely missing columns were detected. The duplicate row was retained because identical recorded attributes do not necessarily indicate an erroneous record. Potential inconsistencies in urine specific gravity and urine albumin were also identified and flagged for review, as recording practices may distort summary statistics and relationships.

Are the classes equally represented?

The classes are almost equally represented. The majority class accounts for 51.14% of rows, while the other class accounts for 48.86%. A classifier always predicting the majority class would achieve 51.14% accuracy, but this is only a baseline, not a validated model result.

where are values concentrated, how wide is the range, and are there skewness, gaps, spikes, or multiple peaks?

The numerical features show varying distributions and ranges. Age and blood pressure concentrate around specific ranges with distinct spikes, while clinical measurements like albumin, urea, and creatinine exhibit heavy right-skewness and baseline concentrations. Histograms help identify concentrations, gaps, and spikes, but these patterns alone do not show differences between chronic kidney disease classes.

which categories are common or rare, and is an unknown category substantial?

Older age groups (60_plus and 40_59) are common, while younger age groups (Under_18 and 18_39) are relatively rare. Binary categories like Red_Blood_Cells, Pus_Cells, and appetite status (Appetite_Good / Appetite_Poor) are well-balanced across the dataset. Specific clinical indicator presence categories like Bacteria_Present and Pus_Cell_Groups_Present are less common (~1,300 to 2,400 rows).

The **unknown** categories (Pus_Cell_Groups_Unknown, Bacteria_Unknown, and Appetite_Unknown) are **not substantial**, representing negligible rows (9, 6, and 1 row respectively).

which measurements have unusual values relative to their distribution?

Based on the IQR rule table and corresponding box plots, here is the concise breakdown of features with unusual values (outliers) relative to their fences:

- **High Outlier Frequency:**
  - **Urine_Sugar**: Highest flagged proportion at **12.01%** (1,808 rows > 500.00).
  - **Serum_Creatinine**: Significant flagged proportion at **10.21%** (1,536 rows > 6.715).
  - **Age**: Has **5.65%** flagged values (850 rows, primarily lower values < 15.0).
- **Moderate to Rare Outlier Frequency:**
  - **Blood_Pressure**: **1.33%** flagged values (200 rows > 147.50).
  - **Blood_Sodium**: **0.35%** flagged values (52 rows < 123.00).
  - **Blood_Urea**: **0.03%** flagged values (4 rows > 196.25).
- **No Outliers Detected (0.00%):**
  - Urine_Specific_Gravity, Urine_Albumin, Random_Blood_Glucose, Blood_Potassium, Hemoglobin, Packed_Cell_Volume, White_Blood_Cell_Count, and Red_Blood_Cell_Count have zero flagged values beyond their 1.5 x IQR fences.

does each class occupy a different part of a measurement's range?

No, neither class occupies a distinct, non-overlapping section of a measurement's range. Instead, both **No CKD** and **CKD** classes span almost identical ranges across all numerical measurements, with considerable overlap:

- **Substantial Range Overlap:** Measurements such as Age, Blood_Pressure, Urine_Specific_Gravity, Blood_Potassium, Hemoglobin, Packed_Cell_Volume, and White_Blood_Cell_Count show nearly identical interquartile ranges (IQRs) and total spans for both groups.
- **Shifted Medians and Spread:**
  - **Random_Blood_Glucose & Blood_Urea:** The median and upper quartile are shifted higher for the **CKD** class, though the overall ranges still overlap.
  - **Urine_Sugar:** The **No CKD** class shows higher upper-whisker values up to 1000, whereas **CKD** values are largely constrained below 250 in this visualization.
  - **Serum_Creatinine & Blood_Sodium:** Both classes share nearly identical IQRs and median values, though individual outliers extend similarly across both groups.

In summary, individual numerical measurements alone do not separate the classes into distinct spatial regions; classification will rely on multi-feature combinations.

How do the average and observed extremes differ between target categories?

Features with Notable Differences in Means

- **Urine_Sugar**: The mean for **No CKD** is substantially higher than for **CKD**.
- **Random_Blood_Glucose**: The mean is visibly higher in the **CKD** category compared to **No CKD**.
- **Blood_Urea & Serum_Creatinine**: Both metrics show noticeably higher average values in the **CKD** class than in the **No CKD** class.
- **Urine_Specific_Gravity**: Average values are slightly higher for **No CKD** compared to **CKD**.

2. Comparison of Observed Extremes (Min & Max)

- **Maximums (Max):**
  - **Urine_Sugar**: Displays a significantly higher maximum value in the **No CKD** category compared to **CKD**.
  - **White_Blood_Cell_Count**: Shows a higher maximum recorded value in the **No CKD** group.
  - **Most Other Features**: Metrics such as Age, Blood_Pressure, Urine_Albumin, Blood_Sodium, Blood_Potassium, Hemoglobin, Packed_Cell_Volume, and Red_Blood_Cell_Count share identical or nearly equal maximum upper limits across both categories.
- **Minimums (Min):**
  - Observed minimum limits (e.g., for Age, Blood_Pressure, Blood_Sodium, Hemoglobin, Packed_Cell_Volume, White_Blood_Cell_Count, and Red_Blood_Cell_Count) are nearly identical for both **CKD** and **No CKD** patients.

While **minimum and maximum boundary extremes are mostly shared** between the two classes, the **averages reveal key clinical distinctions**: the **CKD** group exhibits elevated mean levels of **Random Blood Glucose, Blood Urea, and Serum Creatinine**, while the **No CKD** group shows higher average **Urine Sugar** and slightly higher **Urine Specific Gravity**.

how many rows from each class occur in each category?

1. Categorical Breakdown Overview

- **Age_Group:**
  - **60_plus:** ~3,150 CKD rows vs. ~2,950 No CKD rows.

    - **40_59:** ~2,920 CKD rows vs. ~3,110 No CKD rows.
    - **18_39:** ~670 CKD rows vs. ~1,140 No CKD rows.
    - **Under_18:** ~580 CKD rows vs. ~510 No CKD rows.
- **Binary Features with Strong Class Separation:**
    - **Red_Blood_Cells:** All ~7,350 **CKD** rows fall into category 1, while all ~7,690 **No CKD** rows fall into category 0.
    - **Pus_Cells:** All ~7,350 **CKD** rows fall into category 1, while all ~7,690 **No CKD** rows fall into category 0.
- **Appetite Indicators (Appetite_Poor & Appetite_Good):**
    - **Appetite_Poor (1):** ~5,700 CKD rows vs. ~900 No CKD rows.
    - **Appetite_Poor (0):** ~1,600 CKD rows vs. ~6,800 No CKD rows.
    - **Appetite_Good (1):** ~1,600 CKD rows vs. ~6,800 No CKD rows.
    - **Appetite_Good (0):** ~5,700 CKD rows vs. ~900 No CKD rows.
- **Clinical Finding Indicators (Pus_Cell_Groups & Bacteria):**
    - **Pus_Cell_Groups_Present (1):** ~1,400 CKD rows vs. ~1,000 No CKD rows.
    - **Pus_Cell_Groups_Present (0):** ~5,900 CKD rows vs. ~6,700 No CKD rows.
    - **Pus_Cell_Groups_Absent:** Mirror opposite of Pus_Cell_Groups_Present.
    - **Bacteria_Present (1):** ~780 CKD rows vs. ~530 No CKD rows.
    - **Bacteria_Present (0):** ~6,550 CKD rows vs. ~7,160 No CKD rows.
    - **Bacteria_Absent:** Mirror opposite of Bacteria_Present.
- **Unknown Categories (Pus_Cell_Groups_Unknown, Bacteria_Unknown, Appetite_Unknown):**
    - Category 1 contains negligible/near-zero rows for both classes, with almost all ~7,350 CKD and ~7,690 No CKD rows falling into category 0.

Red_Blood_Cells and Pus_Cells provide near-perfect linear separation between classes, while Appetite_Poor shows a heavy concentration of CKD cases. Features like Bacteria and Pus_Cell_Groups show modest elevation in CKD, and Unknown indicators have virtually no rows.

among rows in a category, what percentage belongs to each target class?

Key Categorical Predictors (Strong Separation)

- **Red_Blood_Cells:**
    - **0**: **99.88%** No CKD | **0.12%** CKD (n=7,689)
    - **1**: **0.22%** No CKD | **99.78%** CKD (n=7,360)
- **Pus_Cells:**
    - **0**: **100.00%** No CKD | **0.00%** CKD (n=7,654)
    - **1**: **0.57%** No CKD | **99.43%** CKD (n=7,395$)
- **Appetite_Poor:**
    - **0**: **80.80%** No CKD | **19.20%** CKD (n=8,426)
    - **1**: **13.41%** No CKD | **86.59%** CKD (n=6,623)

- **Appetite_Good:**
  - **1**: **80.80%** No CKD | **19.20%** CKD (n=8,425)
  - **0**: **13.42%** No CKD | **86.58%** CKD (n=6,624)

2. Age Groups (Age_Group)

- **60_plus**: **48.23%** No CKD | **51.77%** CKD (n=6,115)
- **40_59**: **51.51%** No CKD | **48.49%** CKD (n=6,030)
- **18_39**: **62.75%** No CKD | **37.25%** CKD (n=1,812)
- **Under_18**: **46.15%** No CKD | **53.85%** CKD (n=1,092)

3. Clinical Finding Indicators

- **Pus_Cell_Groups_Present:**
  - **0**: **52.84%** No CKD | **47.16%** CKD (n=12,604)
  - **1**: **42.37%** No CKD | **57.63%** CKD (n=2,445)
- **Bacteria_Present:**
  - **0**: **52.24%** No CKD | **47.76%** CKD (n=13,737)
  - **1**: **39.63%** No CKD | **60.37%** CKD (n=1,312)

4. Unknown Categories (Very Small Group Sizes)

- **Pus_Cell_Groups_Unknown:** Category 1 → **77.78%** No CKD | **22.22%** CKD (n=9)
- **Bacteria_Unknown:** Category 1 → **66.67%** No CKD | **33.33%** CKD (n=6)
- **Appetite_Unknown:** Category 1 → **100.00%** No CKD | **0.00%** CKD (n=1)